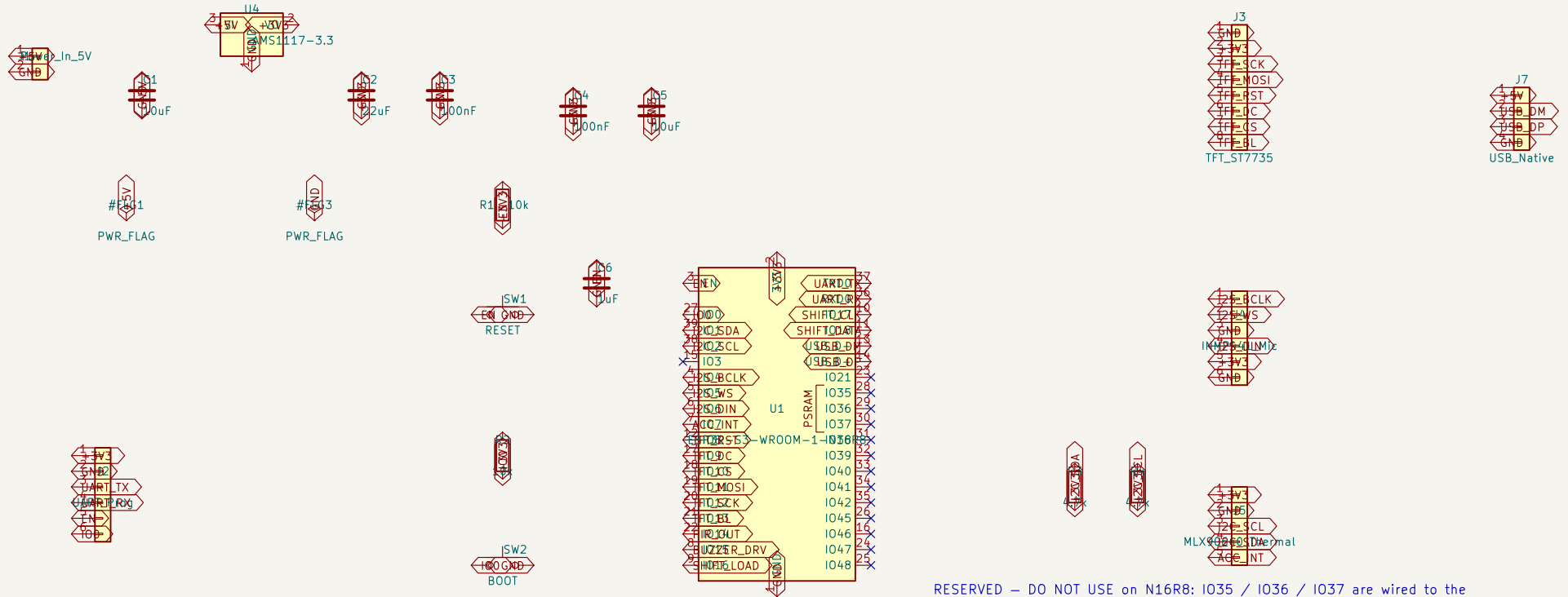


inSomnia – adaptive wake clock | ESP32-S3-WROOM-1-N16R8



KEYPAD – active LOW (10k pull-up per key, switch to GND).
Chain: U2.Q7 -> U3.DS, U3.Q7 -> SHIFT_DATA (U3 is read out first).
74HC165 shifts D7 first, so the firmware bit order is:
bit0..7 = U3 D7..D0 = Key 1..8
bit8..11 = U2 D7..D4 = Key 9..12
U2 D3..D0 tied to +3V3 (spare, always reads "not pressed").

RESERVED – DO NOT USE on N16R8: IO35 / IO36 / IO37 are wired to the octal PSRAM inside the module. IO0, IO3, IO45, IO46 are boot straps. IO46 is input-only. All are left as no-connect here.

J5 = MLX90640 4x4 non-thermal array (I2C 0x33). It sits on the clock aimed at the bed and ~~reads BOTH~~ the mattress accelerometer and the PIR: frame differencing gives contactless actigraphy, and a warm body-sized region gives a direct in-bed / out-of-bed reading. No wire to the bed. Only 4 pins. J5 five pins are used (VIN/GND/SDA/SCL). J6 is not a spare 3-pin header on +5V / IO04 / GND. J4 INMP411 L/R is tied to GND = LEFT channel.

Sheet 1 / File: inSomnia.kicad_sch			
Title:			
Size: 1511	Date: 2024-01-10	SW13	SW14
KICAD 5.0.A. 10.0.6	Key_10	Key_11	Key_12
Rev:			Id: 1/1